

These breaches exposed systemic vulnerabilities in India's telecom infrastructure, including outdated systems and inadequate security practices. The scale of data exposure and service disruptions highlighted the sector's role as a critical national asset, necessitating robust encryption, regular security audits, and proactive threat monitoring. The incidents spurred India to strengthen its cybersecurity framework, including the National Cyber Security Policy, but challenges remain in securing legacy systems against evolving threats.

6. FTX collapse and exchange fraud

The FTX collapse in November 2022 was a monumental financial scam, defrauding customers of \$8.9 billion. Founded by Sam Bankman-Fried (SBF) and Gary Wang, FTX, a centralised crypto exchange, collapsed when SBF's misuse of customer funds for Alameda Research's risky investments was exposed. A liquidity crisis followed, and a suspected inside hack drained \$477 million in cryptocurrencies, likely due to unencrypted private keys. SBF received a 25-year prison sentence and a \$11 billion forfeiture.

Additional examples include the Poly Network hack (2021), where \$600 million was stolen but later returned, and the Binance phishing scam (2023), where fake domains tricked users into leaking wallet keys, costing \$50 million. These incidents, like FTX, exposed centralised exchanges' vulnerabilities, driving demands for proof of reserves, asset segregation, and regulatory oversight. The shift toward decentralised exchanges (DEXs) and self-custodial wallets reflects eroded trust, emphasising accountability in digital finance.

FTX's fallout reshaped the crypto industry, with regulators pushing for transparency and users adopting hardware wallets for enhanced security. The scam's massive scale, amplified by insider betrayal, underscored the need for key management practices and heightened investor vigilance in volatile and rapidly evolving markets.

7. NFT scams

NFT scams surged in the 2020s, exploiting the digital art boom. Globally, hackers hijacked Beeple's Twitter in 2023, posting phishing links for fake NFT giveaways, draining Ethereum and Solana wallets. In India, scammers impersonated Bored Ape Yacht Club, using bots to amplify fraudulent tweets. Victims lost significant sums to phishing sites mimicking OpenSea.

Further examples include the 2022 Azuki scam, where a fake minting site stole over \$2 million in Ethereum, and the 2024 Pudgy Penguins fraud in



India, where scammers used convincing deepfake ads featuring Bollywood stars to lure victims, resulting in losses of \$1.5 million. The 2023 CryptoPunks phishing attack globally tricked users into connecting their wallets to malicious Discord servers, leading to losses of around \$10 million in NFTs.

These scams thrived on social media's open platforms and users' inexperience, exploiting wallet connections to steal private keys. India's low crypto literacy amplified losses, prompting platforms to enhance bot detection and users to adopt hardware wallets. Regulatory calls for clearer NFT guidelines persist, as education remains critical to combat Web3 frauds.

8. India money scams and crypto fraud

India's crypto adoption in the 2020s fueled a wave of scams, exploiting regulatory gaps and low financial literacy. Scammers used Twitter to promote fake giveaways and airdrops, directing users to phishing sites that stole wallet cre-

dentials. Mass tweets tagged thousands, promising high returns, with losses estimated at millions annually.

Examples include the 2021 WazirX impersonation scam, where fake airdrop sites stole \$3 million in crypto assets, and the 2023 "Elon Musk giveaway" fraud, using deepfake videos to trick users into sending \$1.2 million in Bitcoin. The 2022 CoinDCX phishing campaign targeted Indian investors with fake investment apps, draining \$2.5 million. The 2023 "RBI Crypto Fund" hoax lured users into sending \$1 million to fake government-backed wallets.

These frauds leveraged India's growing crypto enthusiasm, using social engineering and deceptive domains to trick users. Regulatory responses included stricter KYC/AML rules and RBI awareness campaigns, but enforcement lags in a decentralised ecosystem. Real-time fraud detection, digital literacy, and international cooperation are essential to curb cross-border scams.

In summary

The 2020s have exposed the dual nature of digital innovation: a world of opportunity riddled with risks. The Twitter hack, SolarWinds attack, LockBit's ransomware, APT36's espionage, and Hathway/BSNL breaches reveal how hackers exploit technical flaws, human errors, and geopolitical tensions. Meanwhile, the FTX collapse, NFT scams, and India's money frauds demonstrate how trust in emerging technologies can be weaponised for profit. These incidents have driven global reforms, stricter regulations, better auditing, and a shift toward decentralised systems, but challenges persist. Organisations and individuals alike must adopt zero-trust models, robust encryption, and proactive monitoring, while individuals need education to navigate scams. The decade's lesson is clear: in a connected world, vigilance is the cornerstone of security. **[d]**